

National Aeronautics and Space Administration

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SUBJECT:

The Light Detection and Ranging (LiDAR) Line-up Assessment for Upcoming Navigation Challenges to Help Ease Risk (LAUNCHER) project (STMD-GCD funded) requires a handheld mobile LiDAR mapping instrument to provide a commercial-off-the-shelf comparison and ground-truth comparison to data collected by Kinematic Navigation and Cartography Knapsack (KNaCK) mobile LiDAR instrument data. Requirements for the requested instruments are: a handheld, LiDAR-based mapping sensor that includes GPS/GNSS capability, includes integrated cameras, has a sensor with ~100 m ranging and 0.5 cm range resolution, utilizes a sophisticated simultaneous localization and mapping (SLAM) processing solution, and a local-machine (i.e. NASA computer) processing software with a perpetual license that does not require cloud-computing resources or purchase of processing credits, data processing software compatible with plug-ins for Esri ArcPro and Agisoft Metashape software. The identified instrument that meets requirements is the GeoCue TrueView GO 132S mapping system. This is a handheld scanning LiDAR sensor and associated compute, power supply, and processing software to allow for comparable mobile 3D mapping to the existing MSFC capability KNaCK system, and which will allow appropriate comparison.

No existing mobile LiDAR Mapping capability exists within the agency that can provide the same required ground truth and comparison